

# U.S. Fiscal Sustainability, Productive Capacity, and the Structural Headwinds to Debt Stabilization

*Resonant Dynamics of Sovereign Debt: Productive Capacity as the Structural Signal*

## ABSTRACT

CBO's February 2026 baseline projects federal debt held by the public rising from 101 percent of GDP in 2026 to 120 percent by 2036, while net interest costs increase from 3.2 percent of GDP in 2025 to 4.6 percent by 2036. Those projections do not imply an imminent sovereign crisis. Still, they reflect a persistent deterioration in the interest-growth differential and a rising probability that the fiscal trajectory becomes a material financial-stability concern — not only domestically, but as a determinant of global dollar-funding conditions. This paper examines three structural drivers that compound that trajectory beyond what conventional budget analysis captures: reserve currency costs, security expenditure asymmetries, and deindustrialization. It then develops a productive-capacity framework as a complement to fiscal consolidation, arguing that durable debt stabilization requires both credible primary-balance adjustment and policies that sustain trend growth strong enough to keep the interest-growth differential manageable. The productive-indispensability concept introduced in Section 4 is operationalized through a four-criterion quantitative scoring methodology and further adjusted for geopolitical fragility — the recognition that supply-chain indispensability concentrated in contested jurisdictions is conditional rather than structural. The Treasury-market transmission channel in Section 5 is calibrated against the 2022–2023 regional banking episode through a duration-based capital-impact framework and extended to the cross-border channels through which a domestic fiscal trajectory becomes an international one: global collateral demand for Treasury securities, the foreign-exchange swap basis, and the structural reliance of foreign banking organizations on U.S. wholesale dollar funding. Drawing on Federal Reserve speeches by Bernanke and Yellen, the 2025 Financial Stability Report, BIS commentary on sovereign-debt risk and cross-currency funding, and IMF work on interest-growth differentials, the paper maps the supervisory implications of elevated sovereign debt into specific transmission channels: valuation risk in securities portfolios, wholesale funding pressure — domestic and cross-border — policy-space compression, and Treasury-market functioning. The central conclusion is that fiscal consolidation and productive-capacity investment are complements rather than substitutes, that treating them as such is a prudential rather than ideological position, and that U.S. debt stabilization functions as much as a foundational input to global financial stability as it does a domestic fiscal exercise.

*Keywords:* U.S. public debt; fiscal sustainability; interest-growth differential; productive capacity; financial stability; Federal Reserve supervision; deindustrialization; reserve currency; Herfindahl-Hirschman Index; duration risk; CET1 capital; global dollar funding; foreign banking organizations; cross-currency basis; geopolitical risk

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## 1. Introduction

On December 29, 2006, the United Kingdom made its final payment on a \$4.34 billion loan contracted in 1945, when Britain emerged from the Second World War carrying sovereign debt exceeding 249 percent of GDP. That sixty-one-year repayment arc is a useful reference point not because the United States faces an analogous near-term constraint, but because it illustrates a mechanism that standard fiscal accounting tends to underweight: the debt-to-GDP ratio is as sensitive to the growth rate of the denominator as to the growth rate of the numerator. Britain's nineteenth-century reduction of public debt from approximately 260 percent of GDP in 1815 to roughly 30 percent by 1900 resulted substantially from the productivity gains of industrialization rather than from austerity (Office for Budget Responsibility, 2023). The same arithmetic, working in reverse, explains how obligations incurred at 249 percent in 1945 required six decades to discharge.

The United States faces a different situation: larger productive reserves, a more entrenched reserve currency position, and no prospect of formal default under foreseeable conditions. But the trajectory of the fiscal accounts warrants supervisory attention on its own terms. CBO's February 2026 baseline projects debt held by the public rising from 101 percent of GDP in 2026 to 120 percent by 2036, while net interest costs increase from 3.2 percent of GDP in 2025 to 4.6 percent by 2036 (Congressional Budget Office, 2026; Committee for a Responsible Federal Budget, 2026). CRFB's summary of that baseline notes that debt would surpass the post-World War II record by FY 2030. The Federal Reserve has repeatedly identified the

long-run fiscal path as a legitimate macro-financial concern. Bernanke emphasized that debt-to-income ratios cannot rise without limit and that fiscal adjustments will occur one way or another; Yellen warned that high and rising debt can restrain capital formation, productivity, and growth (Bernanke, 2010, 2011; Yellen, 2010). More recently, the Federal Reserve's April 2025 Financial Stability Report identified U.S. debt sustainability as one of the most frequently cited risks in market outreach — a ranking consistent with the analytical priorities of this paper (Federal Reserve Board, 2025). Episodes of sharp Treasury yield movement, whatever their proximate cause, serve as reminders that sovereign-financing conditions can interact with broader market functioning in ways that compress the time available for supervisory response. Because the dollar intermediates the large majority of global trade invoicing and cross-border bank funding, those same episodes propagate beyond U.S. borders through channels — global collateral demand, the foreign-exchange swap market, and the balance sheets of the foreign banking organizations operating in the United States — that this paper treats as integral to, rather than separate from, the domestic supervisory analysis.

This paper is written from the perspective of financial-institution supervision rather than partisan budget politics. The analytically relevant question for a Federal Reserve policy analyst is not whether the United States will imminently default. It is whether the present fiscal trajectory raises the probability of higher term premia, fixed-income valuation losses, tighter credit conditions, and reduced policy space in future downturns — domestically, and, through the channels developed below, internationally. Those are prudential questions as much as fiscal ones, and they require attention to the interaction between sovereign borrowing, Treasury-market functioning, bank balance sheets, and the economy's productive capacity.

The paper advances four analytical contributions. First, it identifies three structural headwinds — reserve currency costs, security expenditure asymmetries, and deindustrialization — that have compounded the fiscal gap independently of specific budget decisions and that have no direct counterpart in most advanced-economy fiscal comparisons. Second, it develops a productive-capacity framework connecting trend growth to debt dynamics through the interest-growth differential, arguing that frontier innovation, energy infrastructure, and advanced manufacturing belong in any supervisory analysis of fiscal sustainability — and that the indispensability of the sectors involved must be assessed net of the geopolitical fragility of the jurisdictions in which they are concentrated, not merely on the basis of domestic engineering and capital-investment requirements. Third, it maps the supervisory implications of elevated sovereign debt into specific transmission channels relevant to financial institution oversight. Fourth, it extends those channels across borders, showing how a deteriorating U.S. fiscal trajectory propagates into global collateral markets, the foreign-exchange swap basis, and the funding decisions of foreign banking organizations whose U.S. branches are structural net borrowers of overnight wholesale dollars — converting what could be analyzed as a contained domestic balance-sheet problem into a determinant of dollar-funding conditions in Europe and Asia. The conclusion is not that the United States is on the verge of default, nor that productive investment can substitute for fiscal adjustment. It is that the two are complements, that supervisory analysis should treat them as such, and that the stakes extend beyond U.S. financial institutions to the global financial architecture the dollar underpins.

## 2. Fiscal Sustainability and Debt Arithmetic

Fiscal sustainability, for purposes of this paper, means a debt-to-GDP ratio that is stable or declining over time — the standard formulation used by the IMF, the CBO, and in Federal Reserve speeches by Bernanke (2011). The standard debt dynamics identity provides the core analytical framework:

$$(\text{Debt}/\text{GDP})_t = [(1 + r_t) / (1 + g_t)] \times (\text{Debt}/\text{GDP})_{t-1} + \text{PrimaryDeficit}_t / \text{GDP}_t$$

Where  $r$  is the average nominal interest rate on outstanding government debt and  $g$  is nominal GDP growth. When  $r$  exceeds  $g$  — when the interest-growth differential is positive — the debt ratio tends to rise even when the primary balance is near zero. When  $g$  exceeds  $r$ , debt can stabilize more readily without requiring large primary surpluses. This identity clarifies why the same fiscal deficit can be benign in one macroeconomic environment and destabilizing in another. Policy can affect

both terms: lowering the primary deficit reduces the second term; raising trend growth improves the first term by narrowing  $r-g$ . IMF work on interest-growth differentials in advanced economies emphasizes that  $r-g$  is not a fixed parameter — it is endogenous to policy, demographics, productivity, and market expectations (IMF, 2018). That endogeneity is central to the productive-capacity argument developed in Section 4.

That endogeneity also has a cross-border dimension that domestic debt-arithmetic exercises often suppress. The rate  $r$  is not set in a closed economy. It reflects, in part, the price at which a deep pool of non-U.S. official and private holders is willing to fund the outstanding stock of Treasury debt, and the price at which non-U.S. banks are willing to fund their own dollar books using that same debt as collateral. Section 3.1 develops global demand for Treasuries as the world's primary collateral asset as a structural determinant of  $r$  in its own right, distinct from, but interacting with, the reserve-currency financing-cost benefit discussed there.

An important complication comes from Blanchard's 2019 AEA presidential address, which argues that  $r$  below  $g$  has historically been the norm rather than the exception in advanced economies, and that when  $r < g$  persistently, even large primary deficits may not be indefinitely unsustainable (Blanchard, 2019). That argument deserves direct engagement rather than dismissal. If Blanchard's historical characterization is correct, it complicates the urgency framing of this paper's central argument. Two responses are relevant. First, the current U.S. fiscal trajectory may represent precisely the regime shift away from the low- $r$  environment Blanchard describes: CBO's 2026 baseline projects net interest costs rising from 3.2 to 4.6 percent of GDP over a decade, a trajectory inconsistent with a persistently  $r < g$  regime at the projected debt levels. Second, Blanchard himself acknowledges that the  $r < g$  condition is not guaranteed and that high debt raises the probability of adverse fiscal dynamics if that condition reverses. The productive-capacity framework developed here is, in part, an argument about maintaining the conditions — higher trend  $g$ , broader tax base, sustained external demand for dollar assets — that make Blanchard's more benign scenario more likely to persist.

Current public projections point in the wrong direction on both terms of the identity. CBO's baseline shows deficits averaging 6.1 percent of GDP over the 2026–2036 decade — well above any plausible primary-balance stabilization level (Congressional Budget Office, 2026). Net interest costs are projected to reach 4.6 percent of GDP by 2036, exceeding the historical peacetime share of defense spending and representing the fastest-growing component of the federal budget. Spending reaches 24.4 percent of GDP by 2036 while revenue rises only to 17.8 percent, leaving a structural gap not closed by either the baseline growth assumptions or the current revenue trajectory.

Table 1. Selected Baseline Fiscal Indicators

| Indicator                          | 2025 / 2026 | 2030       | 2036 |
|------------------------------------|-------------|------------|------|
| Debt held by the public (% of GDP) | 101 (2026)  | 106        | 120  |
| Net interest costs (% of GDP)      | 3.2 (2025)  | —          | 4.6  |
| Average deficits (% of GDP)        | —           | 6.1 (avg.) | 6.7  |
| Federal spending (% of GDP)        | 23.1 (2025) | —          | 24.4 |
| Federal revenue (% of GDP)         | 17.2 (2025) | —          | 17.8 |

Source: CBO February 2026 baseline as summarized by CRFB (2026).

Table 2 presents a simplified scenario analysis illustrating how the 10-year debt trajectory responds to variation in the  $r-g$  differential, holding the primary deficit constant at 2.0 percent of GDP — broadly consistent with the structural primary gap implied by CBO's projections when cyclical factors are removed. The formula applied is the standard first-order approximation  $d_{t+1} = d_t \times (1 + r-g) + pd$ , where  $d$  is the debt-to-GDP ratio and  $pd$  is the primary deficit ratio. This is not a forecast; CBO's full dynamic model incorporates year-by-year evolution of both the primary balance and the differential. The purpose is analytic: to isolate the quantitative sensitivity of debt dynamics to the  $r-g$  term in a form that makes the policy arithmetic transparent.

Table 2. Illustrative Debt-to-GDP Trajectories Under Alternative  $r-g$  Scenarios

| Scenario ( $r-g$ )                 | 2026 (Start) | 2031 (Year 5) | 2036 (Year 10) | Change vs. Baseline |
|------------------------------------|--------------|---------------|----------------|---------------------|
| Adverse: $r-g = +2.0\%$            | 101%         | 121.9%        | 145.0%         | +12.5 pp            |
| CBO Baseline: $r-g \approx +1.0\%$ | 101%         | 116.4%        | 132.5%         | —                   |
| Modest improvement: $r-g = +0.5\%$ | 101%         | 113.7%        | 126.6%         | –5.9 pp             |
| Neutral: $r-g = 0.0\%$             | 101%         | 111.0%        | 121.0%         | –11.5 pp            |
| Favorable: $r-g = -0.5\%$          | 101%         | 108.4%        | 115.6%         | –16.9 pp            |

*Note: Simplified illustrative model. Formula:  $d_{t+1} = d_t \times (1 + r-g) + pd$ , with  $pd = 2.0\%$  of GDP held constant and  $d_0 = 101\%$ . CBO's dynamic model incorporates year-by-year changes in the primary balance and  $r-g$ ; this table isolates differential sensitivity only. Source: Author's calculations; CBO (2026) for baseline calibration.*

Table 3 presents the complementary calculation: the primary balance required to fully stabilize the debt ratio at 101 percent of GDP under each scenario, and the implied fiscal adjustment from the current structural primary deficit of approximately 2.0 percent of GDP. At the CBO baseline differential of approximately +1.0 percent, stabilizing the debt ratio requires not merely eliminating the primary deficit but achieving a primary surplus of approximately 1.0 percent of GDP — a combined adjustment of roughly 3.0 percentage points, or approximately \$870 billion annually at current nominal GDP. Narrowing  $r-g$  by 100 basis points through trend growth improvement reduces that required adjustment by an equivalent 1.0 percentage point of GDP, or approximately \$290 billion annually. The two levers reduce different terms of the fiscal identity; both are currently moving in the wrong direction.

Table 3. Primary Balance Required for Debt Stabilization at 101% of GDP

| Scenario ( $r-g$ )                 | Required Primary Balance       | Adjustment from Current Structural Position | Approx. Annual Magnitude |
|------------------------------------|--------------------------------|---------------------------------------------|--------------------------|
| Adverse: $r-g = +2.0\%$            | Surplus of $\approx 2.0\%$ GDP | $\approx 4.0$ pp of GDP                     | ~\$1.2 trillion          |
| CBO Baseline: $r-g \approx +1.0\%$ | Surplus of $\approx 1.0\%$ GDP | $\approx 3.0$ pp of GDP                     | ~\$870 billion           |
| Modest improvement: $r-g = +0.5\%$ | Surplus of $\approx 0.5\%$ GDP | $\approx 2.5$ pp of GDP                     | ~\$725 billion           |
| Neutral: $r-g = 0.0\%$             | Balanced primary budget        | $\approx 2.0$ pp of GDP                     | ~\$580 billion           |
| Favorable: $r-g = -0.5\%$          | Deficit $\leq 0.5\%$ GDP       | $\approx 1.5$ pp of GDP                     | ~\$435 billion           |

*Note: Stabilization condition:  $\text{surplus} = d \times (r-g)$ . Current structural primary deficit estimated at  $\approx 2.0\%$  of GDP (CBO total deficit minus projected average net interest). Dollar magnitudes use ~\$29 trillion 2026 nominal GDP. Source: Author's calculations; CBO (2026).*

These projections do not incorporate the additional spending pressure implied by recent legislative proposals. The CRFB has estimated that enacted and proposed legislation since early 2025 could add materially to the debt path above the CBO baseline. Under a higher-deficit scenario in which annual deficits average roughly one percentage point above the CBO baseline, debt could approach 200 percent of GDP by mid-century — a trajectory that would raise serious questions about sovereign financing conditions well before that endpoint. That scenario is not a forecast; it is a risk boundary that supervisory analysis should keep within view.

#### KEY SUPERVISORY FINDING

The interest-growth differential is the central variable. Under CBO's baseline,  $r$  exceeds  $g$  persistently through the projection window, meaning the debt ratio rises even if primary deficits are held roughly constant. Policies that raise trend growth directly improve  $r-g$  and reduce the primary surplus required for stabilization.

### 3. Three Structural Headwinds

Standard fiscal analysis treats the U.S. deficit primarily as the output of spending and revenue decisions adjustable by Congress. That framing is accurate as far as it goes, but it misses three structural mechanisms that have compounded the deficit independently of specific budget choices and that have no direct counterpart in most advanced-economy comparisons. Supervisory analysis that omits them will systematically underestimate the structural gap between revenues and obligations.

#### 3.1 Reserve Currency Costs

The dollar's reserve currency status confers genuine financial benefits. McKinsey Global Institute (2009) estimated the net annual borrowing-cost benefit at \$40 to \$70 billion — roughly 0.3 to 0.5 percent of GDP — primarily through a 50 to 60 basis point reduction in U.S. Treasury rates. That estimate reflects conditions prevailing before the post-2008 monetary environment and should be treated as broadly indicative of order of magnitude rather than as a current precise figure; the convenience yield on U.S. Treasuries is subject to ongoing empirical estimation and varies with global risk conditions. More recent analysis suggests reserve status increases maximal sustainable debt by approximately 22 percent of GDP (CEPR/VoxEU, 2024). Those advantages are real and directly relevant to the  $r-g$  differential.

They are, however, accompanied by potential offsets that standard fiscal accounting may not fully reflect. Reserve demand has been associated in the literature with exchange rate levels above those that underlying productive competitiveness alone might support — McKinsey Global Institute (2009) places the estimated overvaluation at roughly 5 to 10 percent — which some analysts associate with competitive pressure on tradable goods producers, a dynamic that may have contributed to the structural shift toward financial activity discussed in Section 3.3, though isolating this channel from other determinants of sectoral composition is not straightforward. The dollar's share of allocated global foreign exchange reserves has declined from approximately 71 percent in 1999 to approximately 58 percent by end-2023 per IMF COFER data — a trajectory documented systematically by Arslanalp, Eichengreen, and Simpson-Bell (2022), who attribute part of the decline to active portfolio diversification by reserve managers into nontraditional currencies rather than to passive valuation effects alone. Both the pace of that diversification and its implications for the borrowing-cost premium remain actively debated in the literature.

This reserve-erosion trend can be stated as an explicit transmission mechanism rather than left as a parallel narrative observation. The balance-of-payments identity requires that  $\text{Current Account} + \text{Financial Account} + \text{Capital Account} \approx 0$  (net of errors and omissions), where the current account decomposes as  $\text{CA} = (\text{Exports} - \text{Imports}) + \text{Net Primary Income} + \text{Net Secondary Income}$ . A persistent U.S. current account deficit is, by accounting identity, financed by a financial account surplus — net foreign acquisition of U.S. assets, including Treasuries. As the dollar's reserve share contracts, the same current-account-deficit-driven financing need must increasingly be met by a narrower pool of price-insensitive official reserve managers and a wider pool of price-sensitive private investors. Holding the current account deficit constant, this compositional shift in the financial account requires a higher term premium to clear an unchanged financing need — directly widening the  $r-g$  differential modeled in Section 2.

The borrowing-cost framing above, while standard in the fiscal-sustainability literature, understates the systemic stakes of reserve erosion for a financial-institution supervisor, because it treats the decline as a price effect on U.S. issuance rather than as a structural threat to the global stock of safe collateral. Treasury securities are not merely a financing instrument for the U.S. government; they are the principal high-quality liquid asset against which large parts of the global financial system — foreign central banks, foreign commercial banks, money market funds, and the FX swap market — fund themselves. As of December 2025, foreign holders accounted for approximately \$9.2 trillion, or 31 percent, of the \$30.1 trillion in Treasury debt held by the public, of which 41.9 percent (\$3.9 trillion) was held by foreign official institutions — principally central banks holding Treasuries as the core of their FX reserve and intervention buffers — and 58.1 percent (\$5.4 trillion) by private foreign investors, including the commercial banks that hold Treasuries as high-quality liquid assets under their home

jurisdictions' Basel liquidity rules (Congressional Research Service, 2026). The foreign share of the Treasury market has already fallen from above 50 percent at the peak of the global financial crisis to approximately 30 percent in early 2025, even as the dollar value of foreign holdings has continued to rise in absolute terms (J.P. Morgan, 2026).

This dual role — financing instrument and global collateral asset — is the mechanism through which a deteriorating U.S. fiscal trajectory can generate a global collateral shortage rather than only a domestic financing-cost increase. If term premia widen in response to fiscal deterioration (the mechanism modeled in Section 5.4), the mark-to-market value of the Treasury portfolios held by foreign central banks and foreign commercial banks falls through the same duration mechanics as the domestic bank portfolios analyzed in Table 11. For a foreign central bank, this is a valuation loss on its primary intervention and liquidity buffer. For a foreign commercial bank, it is a valuation loss on the asset it posts as collateral in its own domestic repo and derivatives markets and holds to satisfy liquidity coverage requirements. Both effects tighten the supply of usable collateral at precisely the moment global funding markets are most likely to need it — a mechanism the cross-currency funding literature documented for 2008, when dealers identified an acute shortage of high-quality Treasury collateral as a binding constraint on repo and FX swap market function (Hördahl and King, 2008).

The scale of the underlying exposure is large enough that even a partial collateral impairment has systemic reach. BIS data indicate non-U.S. banks' on-balance-sheet dollar debt totaled approximately \$26 trillion as of mid-2022, with an additional \$39 trillion in off-balance-sheet dollar obligations embedded in FX swaps and forwards — payment obligations that do not appear on balance sheets and that, in combination, exceed the entire on-balance-sheet stock of dollar Treasury bills, repo, and commercial paper (Bank for International Settlements, 2022). This "missing dollar debt" functions as a continuous rollover obligation, collateralized in large part by the same pool of high-quality dollar assets whose supply and price are affected by U.S. fiscal conditions. A widening of U.S. term premia that simultaneously erodes the value of foreign-held Treasury collateral and raises the cost of fresh dollar borrowing compounds, rather than offsets, the stress on this rollover stock.

The transmission channel into FX funding markets operates through the cross-currency basis — the premium non-U.S. institutions pay, relative to covered interest parity, to borrow dollars via the FX swap market rather than directly. BIS research established that this basis, which should theoretically be zero under no-arbitrage conditions, persistently departs from zero in the direction reflecting dollar scarcity and widens further during episodes of dollar funding stress (Borio, McCauley, McGuire, and Sushko, 2016). A decline in foreign demand for dollar assets — whether from active reserve diversification, as documented above, or from a fiscal-risk-driven repricing of the kind modeled in Section 5.4 — reduces the supply of dollars available through the channels that would otherwise keep the basis near parity, pushing non-U.S. institutions toward the FX swap market and widening the basis they pay. In practice, this raises the effective cost of dollar funding for European and Asian banks regardless of the policy rate set by their home central bank — the mechanism that produced acute dollar funding stress in the 2008 crisis and again in March 2020 (Cecchetti and Schoenholtz, 2020).

The policy backstop for this channel is the Federal Reserve's standing U.S. dollar liquidity swap line network with the Bank of Canada, the Bank of England, the Bank of Japan, the European Central Bank, and the Swiss National Bank — converted from temporary to standing arrangements in October 2013 and expanded to nine additional central banks on a temporary basis during the March 2020 dollar funding squeeze (Federal Reserve Board, 2020; Federal Reserve Bank of Richmond, 2024). These swap lines function as a release valve for the mechanism described above: by giving foreign central banks direct access to dollars without forcing private institutions through the FX swap market, they cap the price spike that would otherwise result and reduce the incentive for foreign holders to liquidate dollar assets, including Treasuries, into a stressed market (Brookings Institution, 2025). Their reliability has itself recently become a subject of discussion among foreign officials, following reporting that some foreign central bankers have begun privately asking whether a future Federal Reserve leadership would extend dollar liquidity as readily as it did during the 2008 and 2020 episodes (Brookings Institution, 2025). Whatever the resolution of that question, its emergence is itself a supervisory data point: a channel previously treated as a structural certainty is now, at the margin, a variable — and a fiscal trajectory that erodes confidence

in U.S. assets generally is the kind of background condition under which such questions tend to be asked more, not less, frequently.

Table 4 summarizes the scale of this exposure.

Table 4. Global Dollar Funding and Collateral Exposure

| Metric                                                                   | Value                                                                                                                                                                 | Source                                                                |
|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Foreign holdings of U.S. federal debt (Dec. 2025)                        | \$9.2T (31% of \$30.1T publicly held debt)                                                                                                                            | Congressional Research Service (2026)                                 |
| — of which foreign official (central bank) holdings                      | \$3.9T (41.9% of foreign holdings)                                                                                                                                    | Congressional Research Service (2026)                                 |
| — of which foreign private holdings (incl. commercial banks)             | \$5.4T (58.1% of foreign holdings)                                                                                                                                    | Congressional Research Service (2026)                                 |
| Foreign share of marketable Treasury market                              | ≈30% (early 2025), down from >50% at GFC peak                                                                                                                         | J.P. Morgan (2026)                                                    |
| Non-U.S. banks' on-balance-sheet dollar debt                             | ≈\$26T (mid-2022)                                                                                                                                                     | Bank for International Settlements (2022)                             |
| Non-U.S. banks' off-balance-sheet dollar obligations (FX swaps/forwards) | ≈\$39T (mid-2022); exceeds combined on-balance-sheet stock of dollar T-bills, repo, and commercial paper                                                              | Bank for International Settlements (2022)                             |
| Fed standing dollar swap-line network                                    | Bank of Canada, Bank of England, Bank of Japan, ECB, Swiss National Bank (standing since Oct. 2013); expanded to 9 additional central banks (temporary) in March 2020 | Federal Reserve Board (2020); Federal Reserve Bank of Richmond (2024) |

*Source: as cited in table; figures are the most recent publicly available as of mid-2026 and are order-of-magnitude indicators rather than real-time exposures.*

The net effect of this section's argument is to recast reserve-currency erosion from a borrowing-cost problem into a global-collateral and global-funding problem. The financing-cost benefit estimated by McKinsey Global Institute (2009) and CEPR/VoxEU (2024) measures what the United States gains from supplying the world's primary safe asset. The contribution here is to make explicit what the rest of the world stands to lose — and to transmit back into U.S. funding markets — if that asset's quality is impaired by the trajectory described in Section 2, a linkage Section 5.2 develops further through the foreign banking organizations that intermediate much of this exposure inside the United States, and that Section 5.4 incorporates into the Treasury-market stress scenario directly.

The privilege partially finances the debt while also generating pressures that may weigh on the productive base from which that debt must ultimately be serviced — and, as shown above, generating a collateral and funding dependency running through the global financial system that a purely domestic borrowing-cost framing does not capture. From a supervisory standpoint, an advantage whose long-run magnitude cannot be assumed constant, whose erosion has a stated transmission path into the term premium, and whose downstream effects extend to the balance sheets of foreign central banks, foreign commercial banks, and the foreign banking organizations operating inside the United States, warrants more cautious treatment in fiscal sustainability analysis than a static reading of current financing conditions would suggest.

### 3.2 Security Expenditure Asymmetry

The United States maintains approximately 750 military installations across 80 countries. SIPRI data indicate that U.S. military spending reached \$997 billion in 2024 — 37 percent of total world military expenditure and 66 percent of all NATO spending combined, using SIPRI's methodology that includes military-related spending across federal agencies beyond the Department of Defense (SIPRI, 2025). European NATO members would have required an additional \$663 billion to reach a 5 percent of GDP alliance standard (SIPRI, 2025).

The supervisory observation here is not a political judgment about alliance burden-sharing. It is a structural observation: this level of expenditure constitutes a persistent floor on federal spending that does not respond to the standard fiscal consolidation toolkit. Discretionary non-defense spending represents less than 15 percent of total federal outlays. The structural floor on security expenditure — combined with the mandatory spending growth discussed in Section 6 — means that the categories of spending most accessible to budget negotiators are not, in aggregate, large enough to close the structural gap without addressing either the security cost structure or mandatory programs directly. Table 5 provides a peer comparison that places this floor in quantitative context.

Table 5. Defense Expenditure: United States and Selected Advanced Economies, 2024

| Economy                | Spending (2024, \$B) | % of GDP (approx.) | Gap vs. U.S. (pp) |
|------------------------|----------------------|--------------------|-------------------|
| United States          | 997                  | ~3.4%              | —                 |
| United Kingdom         | 75                   | ~2.3%              | -1.1              |
| France                 | 67                   | ~2.1%              | -1.3              |
| Germany                | 97                   | ~2.1%              | -1.3              |
| Japan                  | 55                   | ~1.0%              | -2.4              |
| Australia              | 33                   | ~2.0%              | -1.4              |
| NATO avg. (excl. U.S.) | —                    | ~1.9%              | -1.5              |

Source: SIPRI (2025). U.S. figure reflects SIPRI's broader methodology including military-related spending across federal agencies beyond DoD (see §3.2). GDP shares approximate; vary with IMF WEO deflators. Note: a 1.5 percentage point premium above the peer average represents approximately \$430 billion annually at current U.S. nominal GDP — a structural fiscal cost that standard consolidation frameworks treat as fixed.

### 3.3 Deindustrialization and the Financialization Transition

Manufacturing's share of U.S. GDP fell from approximately 27 percent in 1950 to around 11 percent today, with the finance, insurance, and real estate sector surpassing manufacturing as a share of domestic production as early as 1991 (Bureau of Economic Analysis, NIPA Table 1.3.5). The most directly traceable fiscal implication of this structural shift operates through the differential tax-revenue elasticity of different forms of economic activity — a relationship the mainstream empirical literature quantifies with reasonable consistency, and one that can be stated as a bounded coefficient rather than only as a directional claim.

Define the tax-revenue elasticity of a sector as  $\epsilon_{\text{sector}} = (\% \Delta \text{Tax Revenue}) / (\% \Delta \text{Sector Value Added})$ . Moretti (2010) estimates a tradable-sector local employment multiplier of 1.6 to 2.0 — each direct job in the tradable sector supports one to two additional positions in the local service economy through income and spending linkages, a relationship that does not hold at the same magnitude for financial-services employment, which is more geographically concentrated and generates narrower local consumption spillovers (Moretti, 2010). Applying the combined OASDI and HI payroll tax rate of 15.3 percent to this multiplier yields a bounded payroll-tax-base capture range:

$$\Delta \text{Payroll Tax Base per \$1 of direct tradable-sector wages} \approx (1 + \text{local multiplier}) \times \text{Wage Share of Value Added} \times 0.153$$

For tradable manufacturing, where compensation is predominantly ordinary wages, this implies approximately **2.6× to 3.0× greater payroll-tax-base capture per dollar of direct wage activity** than an equivalent dollar of activity that does not generate the same induced-employment effect. The comparison to financial intermediation requires a second, independent term. Per Bivens and Mishel (2013), a disproportionate share of financial-sector compensation at the top of the distribution is realized as capital gains, carried interest, or deferred compensation rather than ordinary wages, and is taxed at materially lower effective rates than the 15.3 percent payroll rate applied to wage income. The revenue-elasticity gap between manufacturing and finance is therefore the product of two distinct mechanisms — the employment-multiplier differential and the compensation-composition differential — and should be stated as such rather than collapsed into a single qualitative

sentence. Table 6 summarizes both terms.

Table 6. Tax-Revenue Elasticity and Payroll-Tax-Base Capture, by Sector

| Sector                   | Local Employment Multiplier                             | Wage-Based Comp Share                                                                                                   | Illustrative Payroll-Tax-Base Capture (per \$1 direct wages)                                                                                                                                                       |
|--------------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tradable manufacturing   | 1.6–2.0 (Moretti, 2010)                                 | High — predominantly ordinary wages                                                                                     | $(2.6 \times -3.0 \times) \times 0.153 \approx \$0.40\text{--}\$0.46$                                                                                                                                              |
| Financial intermediation | Lower; more geographically concentrated (Moretti, 2010) | Lower at the top of the distribution — capital gains, carried interest, deferred compensation (Bivens and Mishel, 2013) | Materially below the manufacturing range; a precise sector-specific multiplier is not available in the literature reviewed and should not be asserted to a decimal point absent BEA sector-level compensation data |

*Note: Wage share of value added differs by sector and is not held constant in this formula; a complete elasticity estimate would require sector-specific wage-share data (BEA, NIPA Table 1.10) as a third term. This table isolates the multiplier and compensation-composition channels that the literature supports; it is a structured heuristic, not a fully estimated elasticity model. Source: Author's calculations from Moretti (2010) and Bivens and Mishel (2013).*

The fiscal consequences of manufacturing contraction have been documented systematically by Autor, Dorn, and Hanson (2013), whose analysis of import competition finds that regions experiencing manufacturing job losses exhibit material increases in disability, unemployment, and income-support transfers — raising public expenditures while simultaneously eroding the payroll tax base (Autor, Dorn, and Hanson, 2013). BEA data show that compensation in financial services is more heavily concentrated among highly paid professionals and disproportionately realized as capital gains, carried interest, or deferred compensation — forms subject to lower effective payroll tax rates than ordinary wages (Bivens and Mishel, 2013).

The growth consequence is measurable independently of any hegemonic-cycle framework. U.S. real GDP per capita has grown at roughly 1.5 percent annually over the past fifteen years — below the pace of several comparable advanced economies despite U.S. advantages in technology and research. Federal Reserve research has documented the drag that deindustrialization has imposed on regional labor markets, multiplier effects, and aggregate demand (Federal Reserve Bank of Chicago, 2010). That drag on growth tends to widen the  $r$ – $g$  differential and increase the primary surplus required for debt stabilization through the mechanism quantified above, though the elasticity coefficients in Table 6 are themselves bounded estimates and the aggregate effect operates through several intermediating channels that are difficult to isolate fully from other structural factors. To the extent that financialization is itself partly a product of the reserve currency premium described in Section 3.1 — which overvalues the dollar and disadvantages tradeable goods production — the three headwinds may reinforce each other in ways that no single budget lever can fully address.

## 4. A Productive-Capacity Framework

Traditional discussions of debt sustainability often stop at fiscal consolidation: lower spending, higher taxes, or some combination. Those measures matter, but they are incomplete if the underlying economy is losing productive momentum. Bernanke wrote that public policy should support workforce skills, private capital formation, research and development, and public infrastructure; Yellen emphasized that delay makes adjustment more painful and less controllable (Bernanke, 2011; Yellen, 2010). The debt arithmetic in Section 2 formalizes the same point: any policy that raises trend growth narrows  $r$ – $g$  and reduces the primary surplus required for stabilization.

Productive capacity, as used here, denotes the set of conditions that sustain broad-based output growth over time: high-quality physical infrastructure, reliable energy systems, a capable and accessible labor force, frontier technology, and institutions that support long-term investment. That is a narrower and more defensible concept than the claim that the United States can simply 'grow out' of its debt — a claim this paper does not make. The argument is that stronger productive

capacity improves the probability that debt dynamics remain manageable without requiring abrupt fiscal adjustment, and that the structure of the economy materially affects the burden of public liabilities, as the Geithner (2005) speech on productivity and fiscal sustainability observed directly.

A more specific concept is essential to the argument: productive indispensability. Standard endogenous growth theory (Romer, 1990) establishes that knowledge-intensive sectors generate positive externalities — non-rival ideas that spill over to other producers. Productive indispensability is a more sector-specific analytical claim: it refers to sectors whose outputs function as high-value, difficult-to-substitute inputs to global production systems in ways that are associated with switching costs that may generate sustained sourcing inertia. The distinction matters for fiscal analysis. An industry generating knowledge spillovers raises  $g$  broadly. An industry generating structural dependencies may raise  $g$  while also influencing sovereign financing conditions through channels — including reserve currency demand and the broader conditions under which international economic arrangements are sustained — that do not appear in standard productivity accounting and are difficult to quantify precisely.

The empirical test for productive indispensability is therefore not simply high R&D intensity or TFP contribution, but whether the sector's absence would require other economies to undertake costly restructuring of their production systems around alternatives — a test that advanced semiconductor lithography, foundational AI infrastructure, and GPS-dependent logistics networks plausibly appear more likely to meet than standardized financial intermediation services. As the quantitative matrix below shows, however, "foundational AI infrastructure" is not analytically uniform: the chip-design-and-fabrication layer underlying it exhibits substantially greater concentration than the cloud-services layer through which most economic activity accesses compute, and the two should not be scored as a single category.

A second qualification precedes the matrix itself and bears directly on the fiscal mechanisms claimed below. The geographic location of productive indispensability and the location of its fiscal benefit are not the same variable. A sector can be globally indispensable — meaning its absence would force costly restructuring of world production systems — without that indispensability being domestically sited, in which case the payroll- and corporate-tax-base benefits described later in this section do not accrue to the United States in proportion to the sector's global importance. This distinction is made explicit in the scoring methodology below and should be read as a qualification on, not a rejection of, the productive-indispensability argument: it sharpens which sectors actually deliver the fiscal mechanism this paper describes, as opposed to which sectors are merely globally critical.

A third qualification is geopolitical rather than economic in character, and it precedes the matrix for the same reason. The empirical test above implicitly assumes that a sector's current configuration is stable — that access to it is a function of engineering and capital constraints rather than of contested territorial control or discretionary foreign export licensing. That assumption fails for the two highest-scoring sectors in the matrix below. Advanced-node semiconductor fabrication is concentrated in Taiwan, separated from mainland China by a strait through which approximately 20 percent of global maritime trade and 44 percent of the world's container fleet transit in ordinary conditions, and across which industry risk analysts citing Bloomberg Economics estimates put the global GDP at risk from a sustained military conflict or blockade at on the order of \$10 trillion, or roughly 10 percent of world GDP (Bloomberg Economics, as reported in semiconductor supply-chain risk analysis, 2026). EUV lithography is concentrated in a single Dutch firm whose ability to sell, service, and maintain its most advanced tools is currently subject to an active and unresolved dispute between the U.S. and Dutch governments over the scope of export controls to China — a dispute the Dutch trade minister was still actively lobbying Washington to narrow as of late June 2026 (Bloomberg, 2026; NL Times, 2026). Productive indispensability, in other words, cannot be reduced to a question of what can be manufactured domestically. It is conditional on the continued stability of a small number of fragile bilateral arrangements — between Washington and Taipei, and between Washington and The Hague — that lie outside the control of U.S. fiscal or industrial policy in the near term. The scoring framework below makes this conditionality explicit rather than leaving it as a background assumption.

## 4.1 Quantified Scoring Methodology

Five criteria are each converted to a 0–10 score from a defined metric. Four combine into the weighted composite PI Index used in earlier drafts of this framework; the fifth is applied separately, as a discount, for the reason explained below.

**Global Supply Concentration (GSC)** is measured on two sub-dimensions. Market structure is captured by the Herfindahl-Hirschman Index,  $HHI = \sum (s_i)^2$ , where  $s_i$  is each firm's global market share in percentage points. Standard DOJ/FTC Horizontal Merger Guidelines bands are applied: HHI below 1,500 is unconcentrated (score 1–3); 1,500–2,500 is moderately concentrated (score 4–6); above 2,500 is highly concentrated (score 7–10, scaling toward 10 as HHI approaches its theoretical maximum of 10,000 under monopoly). Separately, U.S./allied production share — the percentage of global output produced domestically or by treaty allies — is tracked because, as shown below, market concentration and U.S. fiscal capture frequently diverge.

**Switching Cost Horizon (SCH)** combines the capital expenditure required to build an alternative supply chain at equivalent capability, normalized against the incumbent's annual sector revenue, with the time required to reach yield or quality parity (not merely first output). Low CapEx-and-time combinations score 1–3; moderate combinations score 4–6; CapEx exceeding twice incumbent annual revenue combined with a replication horizon exceeding seven years scores 7–10.

**Downstream Breadth (DB)** is measured via BEA Input-Output Total Requirements Tables: the count of three-digit NAICS sectors for which the candidate sector constitutes a non-trivial share (above 2 percent) of intermediate input costs, combined with the sector's total-output multiplier. A multiplier above 2.5 or dependence by more than fifteen downstream sectors scores 7–10.

**R&D Intensity (RDI)** is R&D expenditure as a share of sector revenue, benchmarked against the U.S. national R&D-to-GDP ratio of 3.43 percent (NSF/NCSES, 2025) and the federally funded R&D share of GDP, which stood at 0.63 percent in 2024 — its lowest sustained level since the 1950s (NSF/NCSES, 2026). Intensity exceeding twice the national baseline scores 7–10.

The composite index weights concentration and switching cost most heavily, consistent with the empirical test stated above:

$$PI\ Index = 0.30 \times GSC + 0.30 \times SCH + 0.25 \times DB + 0.15 \times RDI$$

**Geopolitical Fragility Discount (GFD)** captures a dimension the four criteria above do not: whether the sector's indispensability is conditional on uninterrupted access to a single contested transit corridor, a single non-allied or contested-sovereignty jurisdiction, or a single foreign government's export-licensing discretion. It is scored 0–10, where 0–3 reflects geographically diversified production with no single chokepoint or licensing dependency; 4–7 reflects concentration in a single stable allied jurisdiction whose export posture is nonetheless subject to active bilateral negotiation; and 8–10 reflects concentration in a jurisdiction exposed to an active territorial dispute, a contested strait, or comparable military risk. GFD does not enter the PI Index as a fifth additive term, because it measures a different property — fragility of access rather than depth of indispensability. It is instead applied as a multiplicative discount:

$$PI\_adjusted = PI\ Index \times (1 - GFD/20)$$

so that a sector scoring at the ceiling of structural indispensability but also at the ceiling of geopolitical fragility is discounted by up to 50 percent, while a sector with no meaningful chokepoint exposure retains its full PI Index score.

## 4.2 Applied Scoring

Table 7. Quantified Productive Indispensability Scoring Matrix

| Sector                                                        | Market HHI                                                                                                                               | U.S./Allied Share                                                                                              | SCH             | DB         | RDI        | PI Index (0–10) | GFD (0–10)                                                                                                       |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------|------------|------------|-----------------|------------------------------------------------------------------------------------------------------------------|
| Advanced-node semiconductor fabrication                       | ≈5,020 (TSMC 69.9%, Samsung 7.2%, SMIC 5.3%, UMC 4.4%, GlobalFoundries 3.9%, other ≈6.8% fragmented)                                     | ≈0% leading-edge domestic capacity at scale (Taiwan-sited); rising via CHIPS-Act-linked fabs not yet at parity | High (9)        | High (9)   | High (8)   | <b>8.4</b>      | <b>9.5</b> — Taiwan Strait transit/conflict exposure; no fast substitution                                       |
| EUV lithography                                               | 10,000 — single supplier, 100% of EUV systems; same firm holds roughly 90%+ of the broader lithography-tool market                       | 0% (Netherlands; NATO-allied, non-U.S.)                                                                        | Very high (10)  | High (9)   | High (8)   | <b>9.3</b>      | <b>5.5</b> — allied jurisdiction, but subject to an active, unresolved U.S.–Dutch export-licensing dispute       |
| AI training / cloud compute infrastructure (hyperscale layer) | ≈1,790 (top three hyperscalers at approximately 31%, 25%, and 12%; remainder fragmented)                                                 | High — the largest hyperscalers are U.S.-domiciled                                                             | Medium (6)      | High (8)   | Medium (6) | <b>6.7</b>      | <b>2.5</b> — low direct chokepoint exposure at the services layer                                                |
| GPS / satellite PNT services                                  | Non-market structure — single sovereign operator of the de facto global civil-timing standard despite nominal constellation alternatives | 100% of the GPS constellation specifically                                                                     | High (8)        | High (7)   | Medium (5) | <b>7.0</b>      | <b>3.0</b> — sovereign-controlled, no single foreign chokepoint, though distinct anti-satellite/jamming exposure |
| Pharma biologics / mRNA platform manufacturing                | Moderately-to-highly concentrated among a small number of GMP-capable CDMOs (illustrative; primary-source verification pending)          | Mixed — meaningful U.S. capacity, with key CDMO capacity also sited in Asia and Europe                         | Medium-high (7) | High (8)   | High (8)   | <b>7.3</b>      | <b>4.5</b> — partial Asia-sited CDMO concentration                                                               |
| Aerospace and defense systems                                 | Concentrated among a small number of prime contractors (illustrative; primary-source verification pending)                               | High — U.S. primes dominate allied procurement                                                                 | High (8)        | Medium (6) | High (8)   | <b>7.1</b>      | <b>3.5</b> — largely allied-sited, with residual exposure via rare earth and specialty component sourcing        |
| Advanced manufacturing (tradable, broad)                      | Low-moderate; fragmented across many firms and countries                                                                                 | Mixed                                                                                                          | Medium (5)      | High (7)   | Medium (5) | <b>5.5</b>      | <b>4.0</b> — mixed geography                                                                                     |
| Financial intermediation services                             | Low — no single firm exceeds low-single-digit global share                                                                               | High (U.S. and allied dominance of wholesale finance)                                                          | Low (2)         | Medium (4) | Low (2)    | <b>2.7</b>      | <b>1.0</b> — geographically diversified                                                                          |

*Note: Ratings combine documented market-structure data with structured analytical judgment; rows flagged "illustrative" require primary-source verification (BioPlan Associates capacity data for biologics CDMOs; SIPRI/Jane's revenue data for defense primes) before this matrix is represented as fully verified. GPS/PNT does not map onto a conventional market-share denominator and is scored as a qualitative-to-quantitative conversion. This matrix is a structured heuristic for comparative supervisory analysis, not a calibrated transmission model. Sources: TrendForce data as reported by Taipei Times (2026) and Tom's Hardware (2026); ASML FY2025 SEC filings; Synergy Research Group (2026); NSF/NCSES*

(2025, 2026); Bloomberg Economics, as reported in semiconductor supply-chain risk analysis (2026); Bloomberg (2026); NL Times (2026); Romer (1990); Mazzucato (2013); SIPRI (2025).

Table 8. Risk-Adjusted Productive Indispensability Ranking

| Sector                                   | PI Index | GFD | Discount Factor (1 – GFD/20) | PI_adjusted | Rank (raw → adjusted) |
|------------------------------------------|----------|-----|------------------------------|-------------|-----------------------|
| EUV lithography                          | 9.3      | 5.5 | 0.725                        | <b>6.74</b> | 1 → 1                 |
| AI / cloud compute (hyperscale layer)    | 6.7      | 2.5 | 0.875                        | <b>5.86</b> | 3 → 2 (tied)          |
| Aerospace and defense systems            | 7.1      | 3.5 | 0.825                        | <b>5.86</b> | 5 → 2 (tied)          |
| GPS / satellite PNT services             | 7.0      | 3.0 | 0.850                        | <b>5.95</b> | 4 → 2                 |
| Pharma biologics / mRNA manufacturing    | 7.3      | 4.5 | 0.775                        | <b>5.66</b> | 6 → 5                 |
| Advanced-node semiconductor fabrication  | 8.4      | 9.5 | 0.525                        | <b>4.41</b> | 2 → 6                 |
| Advanced manufacturing (tradable, broad) | 5.5      | 4.0 | 0.800                        | <b>4.40</b> | 7 → 7                 |
| Financial intermediation services        | 2.7      | 1.0 | 0.950                        | <b>2.57</b> | 8 → 8                 |

Source: Author's calculations from Table 7.

Three findings follow directly from quantification that were not visible under the prior ordinal rating. First, the two highest-scoring raw sectors — advanced-node fabrication and EUV lithography — are sited almost entirely outside the United States; the domestic fiscal mechanism described below in Section 4.3 does not apply to them in their present geographic configuration. Second, the hyperscale cloud-compute layer, at an HHI near 1,790, sits in the "moderately concentrated" band — well below the "highly concentrated" semiconductor layer it depends on, and the indispensability properties commonly attributed to "AI infrastructure" as an undifferentiated category attach more accurately to the upstream chip layer than to the hyperscaler layer through which most economic activity accesses compute. Third, and most consequential for policy: once geopolitical fragility is priced in, advanced-node semiconductor fabrication — the sector most frequently cited as the centerpiece of reshoring policy — falls from the top of the raw ranking to second-to-last, below every sector except diversified financial intermediation. This is not because the sector is any less indispensable in an engineering sense; it is because that indispensability is concentrated in the single most exposed jurisdiction in the entire matrix. EUV lithography, by contrast, retains the top risk-adjusted rank precisely because its fragility runs through an allied-jurisdiction export-licensing dispute rather than a contested strait — a materially different, and on current evidence less acute, category of risk.

#### KEY SUPERVISORY FINDING

Productive indispensability cannot be manufactured at home. The risk-adjusted ranking in Table 8 shows that the sector with the strongest raw case for domestic reshoring — advanced-node fabrication — is also the sector whose effective indispensability is most undermined by a risk the United States cannot resolve through domestic capital allocation alone. Sustaining productive indispensability in this category requires managing the trade, diplomatic, and security alliances — with Taiwan specifically, and with the Netherlands on a different but related axis — on which access to the sector's current configuration depends. A supervisory framework that prices semiconductor indispensability without pricing the geopolitical fragility of its supply chain will systematically overstate the reliability of the fiscal mechanism this section otherwise describes.

### 4.3 Fiscal Mechanisms and Historical Precedent

The current AI investment cycle exhibits several characteristics historically associated with platform-level technology transitions: the infrastructure buildout for large-scale compute — data centers, power systems, specialized chip fabrication,

advanced cooling — spans multiple sectors, creates network effects, and may generate productivity diffusion beyond the immediate application layer. Whether the breadth of that diffusion approaches earlier general-purpose technology waves remains an open empirical question. The geographic dimension is, however, more directly grounded in available fiscal evidence, and is now subject to the qualifications in Sections 4 and 4.2: domestically sited construction, chip fabrication, and engineering employment generate taxable payroll and corporate revenue at each stage of the value chain in ways that supply chains heavily concentrated offshore do not, at least not to the same degree — a distinction relevant to the revenue term of the fiscal identity, and one that, per the matrix above, currently applies more to the cloud/data-center layer of the AI buildout than to the chip-fabrication layer underlying it — itself the layer carrying the highest unadjusted indispensability score and the highest geopolitical discount.

The structural incentive mismatch between private capital markets and platform-level innovation is worth stating precisely, because it directly bears on the fiscal case for public research investment. Venture capital and private equity operate under return horizons and portfolio-construction disciplines that tend to under-incentivize the category of research most relevant to productive indispensability. The median venture fund targets exit within five to eight years; the payoff horizon for foundational technological platforms — the kind that create new industries rather than improve existing products — typically runs fifteen to twenty-five years from initial research to broad economic diffusion. Private capital funds the later commercialization stages once platforms are sufficiently de-risked. It rarely funds the earlier, longer, and riskier research phases that determine whether a platform-level advance becomes available at all. GPS had no private-sector funder before it was operational; the foundational research underlying mRNA vaccines was sustained by NIH grants for decades before any commercial application was apparent. The implication for fiscal policy is not that government should replace private capital in commercialization — it should not — but that the public R&D budget is not functionally equivalent to private R&D spending, and declining it toward 0.63 percent of GDP from its Cold War peak of 1.86 percent leaves a structural gap in the innovation pipeline that private markets are unlikely to close on their own (NSF/NCSES, 2026).

Three fiscal mechanisms follow from productive indispensability specifically, qualified now by both the geographic-versus-fiscal-capture distinction and the geopolitical-fragility discount above. First, manufacturing and technology ecosystems tend to generate broader tax revenues than financial intermediation, per the elasticity coefficients in Table 6: a dollar of manufacturing value added supports substantially more employment, and therefore more payroll, income, and sales tax revenue, than a dollar of financial activity, improving the revenue term of the fiscal identity, all else equal — provided that activity is both domestically sited and not contingent on an unresolved geopolitical chokepoint. Second, there is historical evidence that productive leadership has been associated with a broader set of instruments for sustaining international institutional commitments — an association that, while difficult to isolate in a formal accounting framework, may bear on the non-market components of sovereign financing conditions over longer horizons. Third, productive leadership appears to be associated with relatively stable international demand for a nation's bonds and currency, a pattern consistent with Eichengreen's historical analysis. The dollar's reserve status reflects both the institutional architecture established at Bretton Woods and the productivity of the American economy that made dollar-denominated assets attractive to hold. Eichengreen's analysis suggests that financial privilege and productive leadership have tended to develop together rather than the former systematically preceding the latter (Eichengreen, 2011) — a pattern consistent with the complementarity argument of this paper.

The historical record supports the basic intuition, though it should be read with care given the institutional and monetary differences across cases. Britain's reduction of public debt from approximately 260 percent of GDP in 1815 to roughly 30 percent by 1900 resulted primarily from industrialization rather than from primary-surplus discipline (Office for Budget Responsibility, 2023). The postwar American experience illustrates the mechanism more directly: debt at 106 percent of GDP in 1945 was reduced substantially through sustained economic expansion associated with broad-based productivity gains and the expansion of American industrial and technological capacity, not through austerity alone. Historical episodes in which productive capacity eroded ahead of sovereign financing conditions — including the Dutch Republic in the eighteenth century — are suggestive, though the monetary structures and institutional contexts differ enough from a modern

fiat reserve currency issuer to preclude direct analogy (de Vries and van der Woude, 1997). Productive leadership in sectors with high spillovers can reinforce long-run trend growth and support the external demand that helps keep sovereign financing conditions stable. This is not a substitute for fiscal discipline. It is a complement to it, and the two should be designed in parallel rather than sequentially.

## 5. Financial Stability and Supervisory Implications

For supervisory purposes, the analytically relevant question is transmission: how does an elevated sovereign debt trajectory translate into risks for financial institutions? High sovereign debt does not become a banking crisis automatically. The pathway runs through specific channels that supervisors can monitor, stress-test, and incorporate into institution-level risk assessment. Four channels warrant particular attention; each, as developed below, has both a domestic and a cross-border dimension.

### 5.1 Duration and Valuation Risk in Securities Portfolios

Large banking organizations and many non-bank intermediaries hold significant quantities of Treasury and agency securities. These holdings serve important liquidity and collateral functions but create duration exposure. In a higher-yield environment — a plausible scenario if term premia reprice in response to fiscal deterioration or a loss of fiscal credibility — unrealized losses accumulate even when credit quality remains strong. The Federal Reserve's April 2025 Financial Stability Report states explicitly that debt-servicing costs for governments and corporations would rise if interest rates increased and that those dynamics could produce fair-value losses on fixed-rate securities among financial intermediaries, reducing credit supply (Federal Reserve Board, 2025). The 2022–2023 experience with regional banking stress under rapid rate increases illustrates that this transmission channel is real, operationally significant, and capable of moving more quickly than supervisory cycle times. A sovereign-risk-driven repricing would differ in origin but not in mechanism; Section 5.4 calibrates this channel against that episode directly and quantifies the resulting capital impact. Section 3.1 shows that the same duration mechanics apply, in parallel, to the Treasury holdings of foreign central banks and foreign commercial banks, meaning the domestic valuation channel modeled here has a foreign-balance-sheet mirror that this paper treats as part of the same transmission event rather than a separate one.

### 5.2 Wholesale Funding Pressure

Deteriorating fiscal confidence, even short of a market event, can increase wholesale funding costs for institutions with significant reliance on short-term liabilities. This is particularly relevant for non-bank financial intermediaries — money market funds, repo-market participants, and others whose structural position in Treasury markets has grown materially over the past decade. The BIS has framed this in directly relevant terms: high government debt combined with the growing presence of non-bank financial institutions in sovereign bond markets creates policy challenges that cut across fiscal, monetary, and prudential domains (Bank for International Settlements, 2025).

This domestic framing, however, omits the institutions most structurally exposed to exactly this channel: the U.S. branches and agencies of foreign banking organizations (FBOs). Most FBO branches lack access to FDIC-insured deposits and consequently depend on wholesale funding — large time deposits, repo, commercial paper, and interbank borrowing — for nearly all of their dollar needs (Federal Reserve Bank of Philadelphia, 2019). This structural feature makes FBOs, in the language of New York Fed research, "marginal price setters" of the price of dollar liquidity in wholesale funding markets, and positions them as the principal bridge between onshore U.S. dollar funding markets and the offshore dollar markets in Europe and Asia described in Section 3.1 (Du, 2023). Historically, FBO branches have operated a "funding branch" model in which the U.S. branch borrows large quantities of wholesale dollars and upstreams them to the parent organization; branches operating on this model rose from holding 40 percent of FBO branch assets in the mid-1990s to 75 percent by 2009, and foreign banks collectively moved from net recipients of parent funding in 1999 to net providers of more than \$600

billion to non-U.S. affiliates by 2008 (Tarullo, 2014). In ordinary times, the U.S. dollar wholesale funding market is not merely a domestic financing venue for FBOs; it is a source of dollars that FBOs distribute to their global organizations.

That structure reverses, predictably, during stress, and the reversal is the transmission mechanism most relevant to the fiscal-risk scenario this paper models. During the 2011 European sovereign debt crisis, U.S. money market funds — proximate, risk-averse investors with no appetite for European sovereign exposure even by association — sharply reduced their funding of European banks' U.S. branches. European parent banks responded by initiating internal liquidity transfers to their U.S. branches to offset the shortfall, and the U.S. branches of euro-area banks became net borrowers vis-à-vis their own head offices in the second half of 2011 for the first time in several years (Federal Reserve Board, International Finance Discussion Papers No. 1059, 2012). The parents absorbed the cost: their own dollar funding costs rose at the same time, since they had to source the dollars to inject — whether through their own wholesale operations or, per Section 3.1, through the FX swap market at a widened cross-currency basis — preventing the liquidity shortfall at the U.S. branch from being offset completely.

The fiscal-risk scenario in this paper runs the same mechanism with the shock originating on the U.S. side rather than the European side. A sharp repricing of Treasury-market term premium — the 200 basis point shock modeled in Section 5.4 — raises wholesale dollar funding costs in the United States directly, through the Tier 1 and Tier 2 collateral-spread widening quantified in Table 12. FBO branches, as structural net borrowers of overnight wholesale dollars with no FDIC-insured deposit base to fall back on, are first in line to absorb that repricing. As in 2011, the likely institutional response is for the parent organization — typically headquartered in London, Frankfurt, Tokyo, or another major financial center — to mobilize home-currency liquidity and swap it into dollars to backstop the New York or Washington branch, rather than allow the branch to curtail its U.S. dollar lending and risk reputational and regulatory consequences in its most important foreign market. That mobilization draws on the parent's home-market liquidity and balance sheet capacity and is transacted, in large part, through the same FX swap market whose basis is already widening under the conditions described in Section 3.1 — meaning the domestic and cross-border channels reinforce rather than operate independently of each other. A U.S. Treasury-market event that might otherwise be analyzed as a contained domestic balance-sheet problem is, through this channel, transmitted directly into the funding markets and balance sheets of foreign parent banks, at the precise moment those parent banks are least able to absorb additional stress cheaply.

The scale of this exposure historically has been large enough to matter at the system level. As of December 2013 — a snapshot from the post-crisis period when FBO balance sheets were unusually cash-heavy following Dodd-Frank implementation — foreign bank branches held approximately 43 percent of total reserves at the Federal Reserve, despite FBO total U.S. assets being a fraction of domestic banks' (Bank for International Settlements, 2014). That figure is dated and should not be read as a current estimate, but it illustrates the order of magnitude of FBO presence inside the U.S. wholesale funding and reserves system; the FFIEC 002 reports that track current branch-level assets, liabilities, and funding composition provide the appropriate basis for an updated, precise calibration of this channel. Table 9 summarizes the transmission sequence.

Table 9. FBO Cross-Border Liquidity Transmission Channel

| Stage                  | Mechanism                                                                                                                              | Empirical Precedent                                                                                                                                                                                             |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Domestic shock      | U.S. Treasury-market term-premium repricing raises wholesale dollar funding costs (Table 12, Tier 1–2 spreads)                         | 200 bps fiscal-stress scenario, Section 5.4                                                                                                                                                                     |
| 2. FBO branch exposure | FBO branches, lacking FDIC-insured deposits, absorb the repricing directly as structural net borrowers of wholesale dollars            | FBOs are "marginal price setters" of wholesale dollar liquidity (Du, 2023)                                                                                                                                      |
| 3. Parent response     | Parent mobilizes home-currency liquidity and swaps it into dollars to backstop the U.S. branch rather than curtail U.S. dollar lending | European parent banks transferred liquidity to U.S. branches in H2 2011; U.S. branches of euro-area banks became net borrowers vis-à-vis head offices for the first time in years (Federal Reserve Board, 2012) |

| Stage                        | Mechanism                                                                                                                                                                     | Empirical Precedent                                                                                                                           |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 4. Cross-border transmission | Parent's dollar-sourcing operation widens the cross-currency basis further (Section 3.1), raising funding costs for the parent's entire global book, not only its U.S. branch | Parent dollar funding costs rose in tandem with the 2011 transfer, preventing full offset of the U.S. shortfall (Federal Reserve Board, 2012) |
| 5. System-level reach        | FBO branches represent a historically large share of dollar liquidity intermediated inside the United States; current aggregate scale requires FFIEC 002 verification         | FBO branches held ≈43% of total reserves at the Federal Reserve as of Dec. 2013 (Bank for International Settlements, 2014)                    |

Source: as cited in table.

### 5.3 Policy-Space Compression

The most diffuse but potentially most consequential supervisory implication is the progressive compression of policy space available in future downturns. When fiscal consolidation is deferred until a stress event, the adjustment tends to be abrupt — with larger multiplier effects on output than a gradual adjustment would produce. Blanchard and Leigh found that fiscal multipliers in the post-2008 period ran approximately 1.5, meaning that for every dollar cut in a high-stress environment, economic output fell by roughly \$1.50 (Blanchard and Leigh, 2013). Bernanke noted this dynamic directly: adjustments occurring in an orderly and gradual way are considerably better than those occurring abruptly in response to a market crisis (Bernanke, 2011). A financial system operating against a backdrop of compressed fiscal space is structurally more fragile than one with room for countercyclical response — the Federal Reserve's own monetary policy transmission is affected by the degree to which fiscal policy can amplify or dampen its actions.

### 5.4 Treasury-Market Functioning: A Quantitative Stress-Testing Framework

The Federal Reserve's 2026 stress-test scenarios assume severe global recession and large asset-price shocks (Federal Reserve Board, 2026). Those scenarios are not sovereign-debt stress tests, but they provide a useful benchmark for the speed at which market conditions can deteriorate. A supervisory scenario focused on sovereign risk would not require assuming default; it would only need to examine whether a sharper repricing of fiscal risk, a more abrupt yield increase, or a loss of market confidence could worsen bank capital and liquidity outcomes under otherwise comparable macro conditions. This subsection operationalizes that scenario, calibrated against the 2022–2023 regional banking episode and expressed as a duration-based capital-impact model, then extends the aggregation to the global channels developed in Sections 3.1 and 5.2.

**Empirical baseline.** SVB Financial Group's HTM securities portfolio carried a weighted-average duration of 6.2 years as of December 2022, and the firm's own disclosed sensitivity implied an approximately 5.6 percent decline in portfolio value per 100 basis point parallel increase in rates across maturities (Federal Reserve Board, 2023; Money, Banking and Financial Markets, 2023). SVB's unrealized losses on HTM and AFS securities combined rose from approximately \$1.6 billion at year-end 2021 to approximately \$17.7 billion at year-end 2022 (Federal Reserve, Office of Inspector General, 2023). The March 2023 forced sale of \$21 billion in AFS securities realized an 8.6 percent loss rate, and the resulting confidence shock produced a \$40 billion deposit run with an additional \$100 billion in unmet withdrawal requests (Federal Reserve, Office of Inspector General, 2023) — the proximate cause of failure was the funding-market run, not the capital-ratio deterioration in isolation. At the system level, aggregate unrealized losses on AFS and HTM securities across FDIC-insured depositories stood at \$478 billion at year-end 2023 (\$204 billion AFS, \$274 billion HTM) and \$481 billion at year-end 2024 — equal to 8.6 percent of the fair value of aggregate securities holdings and 19.9 percent of aggregate banking-subsidary equity (Congressional Research Service, 2024; Office of Financial Research, 2025). Table 10 summarizes this baseline.

Table 10. Empirical Baseline — SVB and System-Wide Unrealized Securities Losses, 2021–2024

| Metric                                        | Value                                          |
|-----------------------------------------------|------------------------------------------------|
| SVB HTM weighted-average duration (Dec. 2022) | 6.2 years                                      |
| SVB portfolio sensitivity                     | ≈5.6% value decline per 100 bps parallel shift |

| Metric                                            | Value                                                                                           |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------|
| SVB unrealized losses, YE2021                     | ≈\$1.6B                                                                                         |
| SVB unrealized losses, YE2022                     | ≈\$17.7B (\$15.2B HTM + \$2.5B AFS)                                                             |
| SVB AFS fire-sale realized loss rate (March 2023) | ≈8.6%                                                                                           |
| Deposit run                                       | \$40B realized; \$100B additional unmet requests                                                |
| System-wide unrealized losses, YE2023             | \$478B (\$204B AFS + \$274B HTM)                                                                |
| System-wide unrealized losses, YE2024             | \$481B = 8.6% of aggregate securities fair value = 19.9% of aggregate banking-subsidiary equity |

Source: Federal Reserve Board (2023); Federal Reserve, Office of Inspector General (2023); Congressional Research Service (2024); Office of Financial Research (2025).

Dividing the OFR's disclosed dollar loss by its disclosed percentage recovers the implied aggregate fair value of FDIC-insured depositories' securities holdings:  $\$481\text{B} / 0.086 \approx \$5.59$  trillion. This figure is not separately published; it is an arithmetic derivation used as the systemic base for the aggregation below.

**Shock calibration and duration formula.** The modeled shock is a 200 basis point parallel upward repricing of term premium, driven by fiscal-risk reassessment rather than monetary tightening, consistent with this section's framing that the transmission mechanism is independent of the cause of the rate move. As of June 25, 2026, the 10-year Treasury yield closed at approximately 4.41 percent against a newly hawkish Federal Reserve policy stance; a 200 basis point move from that level is severe but within the range observed during 2022–2023, not a tail-extreme assumption. The first-order bond-price approximation — consistent with the first-order simplification already used in Tables 2 and 3 — is:

$$\Delta P/P \approx -D_{\text{mod}} \times \Delta y$$

Applying this to SVB's own 6.2-year duration at a 200 basis point shock yields  $\Delta P/P \approx -12.4$  percent, against an empirically observed sensitivity of approximately  $-11.2$  percent (linearly extrapolating the disclosed 100 basis point figure). The roughly 1.2 percentage point gap reflects convexity and AFS/HTM compositional differences and is small enough that the first-order approximation is serviceable for the worked example below.

**CET1 impact — worked example.** Consider an illustrative regional institution, constructed for transmission mapping rather than drawn from any specific filing: total assets of \$50 billion; risk-weighted assets of \$32 billion; CET1 capital of \$3.2 billion (a 10.0 percent CET1 ratio); an AFS-plus-HTM securities portfolio of \$14 billion (28 percent of assets); and a weighted-average portfolio duration of 5.5 years. A 200 basis point shock produces an unrealized loss of  $\$14\text{B} \times 5.5 \times 0.02 = \$1.54\text{B}$ . The capital impact differs by AOCI treatment.

Table 11. Illustrative CET1 Impact of a 200 bps Term-Premium Shock, by AOCI Treatment

| Treatment                                                                                                                          | Mechanism                                                                                                                                                                            | Adjusted CET1                                       | Adjusted CET1 Ratio  | Regulatory Consequence                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A — AOCI opt-out (current treatment for institutions below the proposed \$100B threshold)                                          | Only HTM losses recognized in the proxy metric, tax-adjusted at 24%; assume 60/40 HTM/AFS split, HTM loss = $\$8.4\text{B} \times 5.5 \times 0.02 \times (1-0.24) = \$0.702\text{B}$ | $\$3.2\text{B} - \$0.702\text{B} = \$2.498\text{B}$ | 7.81% (down 219 bps) | Remains above the 7.0% combined capital-conservation-buffer threshold and the 6.5% PCA "well capitalized" threshold; no automatic restrictions                                                                                                                                                                                                        |
| B — Full AOCI inclusion (institutions above the proposed threshold, or under existing rules for the largest banking organizations) | Full pre-tax loss flows through: \$1.54B                                                                                                                                             | $\$3.2\text{B} - \$1.54\text{B} = \$1.66\text{B}$   | 5.19% (down 481 bps) | Falls below the 6.5% PCA "well capitalized" threshold (reclassified "adequately capitalized," triggering brokered-deposit restrictions absent an FDIC waiver) and below the 7.0% combined buffer (triggering the most severe scaled restrictions on capital distributions and discretionary bonus payments); remains above the 4.5% statutory minimum |

*Note: Illustrative balance sheet, not drawn from a specific institution's filings. Tax-adjustment and HTM/CET1 methodology follows published practice for this class of proxy capital-erosion analysis. Source: Author's calculations.*

**Wholesale funding pressure coefficients.** The duration channel above is a quarterly-disclosure phenomenon; the funding channel operates on a faster clock and is empirically the more proximate trigger of failure, per the SVB sequence in Table 10 and the cross-border sequence in Table 9. Table 12 sets out collateral-tier-specific coefficients calibrated against documented stress episodes.

Table 12. Wholesale Funding Pressure Coefficients, by Collateral Tier

| Collateral Tier                              | Repo Rate Spread Widening (bps above EFR, stress episode) | Haircut Adjustment                            | Empirical Precedent                                                                                                                                                                                                            |
|----------------------------------------------|-----------------------------------------------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tier 1 — on-the-run U.S. Treasuries          | +150 to +300+ bps intraday                                | ≈0 pp (historically stable even through 2008) | September 2019 repo spike: intraday rates rose more than 300 bps above the federal funds target upper bound (Office of Financial Research, 2023)                                                                               |
| Tier 2 — off-the-run Treasuries / agency MBS | +50 to +150 bps, sustained                                | +2 to +5 pp                                   | 2020 dislocation-period haircut drift on agency collateral                                                                                                                                                                     |
| Tier 3 — non-agency / lower-rated collateral | +100 to +300+ bps                                         | +10 to +20 pp                                 | 2008 GFC bilateral repo market; haircuts on asset-backed and CLO collateral rose sharply enough that some collateral types ceased to be accepted (Gorton and Metrick, as documented in Federal Reserve Bank of New York, 2012) |

*Source: Office of Financial Research (2023); Federal Reserve Bank of New York (2012).*

A fiscal-risk-driven repricing event propagates through Tier 1 collateral primarily via rate-spread widening with minimal haircut change — meaning the BIS-identified risk to non-bank intermediaries (Section 5.2) operates chiefly through funding-cost pass-through rather than collateral devaluation domestically, even as the parallel cross-currency basis channel (Section 3.1) raises the cost of the dollars that foreign parents must source to backstop their U.S. branches (Section 5.2, Table 9).

**System-wide aggregation.** Applying the duration formula at the system level, using an illustrative system-wide average securities duration of 4.5 years — shorter than SVB's 6.2-year pre-crisis outlier, reflecting post-2023 portfolio repositioning, and itself an input requiring FFIEC Call Report / FR Y-9C verification for precision — a 200 basis point shock implies an incremental systemic loss of  $\$5.59T \times 4.5 \times 0.02 \approx \$503B$ . Added to the existing \$481 billion (YE2024) stock, cumulative unrealized losses under this scenario would approach approximately \$984 billion, roughly double the current level. Scaling the OFR's reported loss-to-equity ratio linearly ( $19.9\% \times [984/481]$ ) implies a loss-to-equity ratio approaching approximately 41 percent — a level without precedent in the OFR's post-2022 reporting series. This is a linear, first-order approximation that does not account for hedging, balance-sheet repositioning, or capital actions banks would take in practice; it functions as a stress indicator of channel severity rather than a point forecast.

This domestic aggregation is, in practice, a lower bound. Section 3.1 showed that foreign central banks and foreign commercial banks hold a substantial share of the same Treasury stock as a primary collateral and liquidity buffer, and that the duration mechanics quantified above apply to their holdings in the same way. A term-premium shock of the kind modeled here therefore generates a parallel valuation loss on foreign-held Treasury collateral at the same time it tightens FBO funding conditions inside the United States (Section 5.2) — and historically, the response of foreign holders to a U.S. dollar liquidity squeeze has been to sell Treasuries to raise dollars rather than to hold them through the stress, a pattern documented in the March 2020 "dash for cash," when foreign official and private selling was a material contributor to that period's Treasury-market dysfunction (Vissing-Jorgensen, 2021; He, Nagel, and Song, 2022). A fiscal-stress event severe

enough to trigger the domestic channels in Tables 10 through 12 is therefore plausibly self-reinforcing at the global level: Treasury-market stress erodes foreign collateral value and tightens FBO funding, and the foreign and FBO response to that tightening — selling Treasuries to raise dollars — adds further selling pressure to the same market whose dysfunction triggered the episode.

Incorporating a fiscal-stress scenario of this kind into the published stress-testing framework — alongside the current global-recession and market-shock scenarios, and alongside the cross-border channels in Sections 3.1 and 5.2 — would bring the Federal Reserve's stress architecture into closer alignment with the risks its own Financial Stability Report has identified as salient.

Table 13. Supervisory Transmission Channels

| Channel                                        | Mechanism                                                                                                                                                                                                                       | Supervisory Relevance                                                                                                                                                              |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Duration / valuation risk                      | Yield increases produce fair-value losses on fixed-rate securities holdings; a 200 bps shock against a representative regional balance sheet erodes CET1 by 219–481 bps depending on AOCI treatment (Table 11)                  | Capital erosion, credit-supply contraction; precedent from 2022–2023                                                                                                               |
| Wholesale funding pressure (domestic)          | Fiscal stress raises repo and money-market funding costs; Treasury-collateral spreads can widen 150–300+ bps intraday with minimal haircut change, while lower-tier collateral sees haircuts rise 10–20 pp (Table 12)           | Liquidity risk for liability-sensitive institutions and non-bank intermediaries; historically the proximate trigger that converts unrealized losses into realized ones             |
| Cross-border FBO funding transmission          | FBO branches, as structural net borrowers of wholesale dollars, force parent banks to mobilize home-currency liquidity and swap it into dollars, exporting the funding shock abroad (Table 9)                                   | Converts a domestic fiscal-stress event into a determinant of foreign parent banks' funding costs and balance-sheet capacity                                                       |
| Global collateral / foreign Treasury valuation | The same duration losses apply to foreign central bank and commercial bank Treasury holdings, tightening global collateral and widening the cross-currency basis (Section 3.1, Table 4)                                         | Raises dollar funding costs in Europe and Asia independent of the home central bank's policy rate; can trigger foreign Treasury selling that compounds domestic market dysfunction |
| Policy-space compression                       | High debt reduces countercyclical fiscal capacity in downturns                                                                                                                                                                  | Amplifies monetary policy transmission risk; limits government backstop capacity                                                                                                   |
| Treasury-market functioning                    | Confidence loss or term-premium repricing disrupts collateral and liquidity; system-wide aggregation implies unrealized losses could approach \$1 trillion under a 200 bps shock, before the foreign-holder feedback loop above | Systemic; affects all institutions using Treasuries as liquidity buffers, domestic and foreign                                                                                     |

Source: Framework drawing on Federal Reserve Board (2025), BIS (2025), IMF (2018), and the stress-testing calibration in Section 5.4.

#### KEY SUPERVISORY FINDING

The duration/CET1 channel and the wholesale-funding channel operate on different clocks, and the cross-border extension of each operates on a third. The CET1 channel is a quarterly-disclosure phenomenon that, per the worked example in Table 11, erodes capital ratios meaningfully but does not by itself produce insolvency for an adequately capitalized institution even under a 200 bps shock. The domestic funding channel is an intraday-to-multiday phenomenon that can force the asset sales converting unrealized losses into realized ones — this was the actual proximate mechanism in the SVB failure. The cross-border channel adds a fourth clock again: it is the speed at which a foreign parent bank can mobilize home-market liquidity and access the FX swap market, which the 2011 precedent suggests is fast enough to transmit a funding shock internationally within the same stress episode, not as a delayed second-order effect. Supervisory analysis that treats the domestic and cross-border channels as sequential rather than concurrent will misjudge how quickly a U.S. fiscal-stress event becomes a global one.

## 6. Policy Options

The relevant policy conclusion is that the United States requires a coherent mixed strategy in which fiscal consolidation and productivity-enhancing investment are treated as complements rather than alternatives. Bernanke's formulation remains analytically useful: a more productive economy eases the tradeoffs, but it does not eliminate the need for fiscal adjustment (Bernanke, 2011). Each lever carries its own costs and timing profile; the question is not which lever is always superior, but whether the policy mix is coherent over time and consistent with the Federal Reserve's supervisory mandate — domestically and, per Section 5, internationally.

### 6.1 Entitlement and Health-Care Adjustment

Mandatory spending on Social Security, Medicare, and Medicaid constitutes roughly 60 percent of the federal budget and grows automatically with demographics and medical cost inflation, without annual appropriations votes. This is the primary structural driver of the long-run debt path. Discretionary non-defense spending — the category most accessible in annual budget negotiations — represents less than 15 percent of total federal outlays. Addressing the long-run debt trajectory without engaging mandatory spending is, as a matter of arithmetic, not possible at the required scale. Entitlement reform is necessary; it is also the category where near-term adjustment has the largest aggregate effect on long-run fiscal sustainability.

### 6.2 Revenue Measures

The United States collects significantly less revenue as a share of GDP than most comparable advanced economies. As Saez and Zucman document, the distributional structure of American taxation has shifted substantially over the past four decades (Saez and Zucman, 2019). Broad-based revenue measures, designed to minimize growth distortions, can reduce annual deficits directly and improve the primary balance term of the debt identity. The net impact on debt dynamics depends critically on the extent to which the revenue structure supports or suppresses productive investment — a revenue system that concentrates the burden on capital formation could offset some of the debt-stabilization gains by further suppressing the denominator. The Section 3.1 balance-of-payments decomposition is relevant here as well: revenue measures that affect the attractiveness of dollar-denominated assets to foreign holders interact with the financial-account financing channel, the global collateral channel, and the cross-currency basis that together help determine the term premium embedded in  $r$ .

### 6.3 Productivity-Enhancing Public Investment

Federal R&D spending has declined from approximately 1.86 percent of GDP in 1964 — during the Cold War space and defense investment programs — to 0.63 percent in 2024, its lowest sustained level since before that peak (NSF/NCSSES, 2026). Private R&D has partially filled the gap, but private capital concentrates heavily on near-term commercialization. Platform-level advances that create new industries rather than improving existing ones — the category most relevant to productive indispensability — typically require the patient, long-horizon research that private capital structures do not efficiently incentivize. The internet did not originate from a venture capital portfolio. Neither did GPS nor the Human Genome Project. The federal government's role is not to select commercial winners; it is to fund the research frontier from which commercially valuable output emerges. The energy grid, immigration pathways for STEM talent, and university-industry research pipelines function as productive infrastructure rather than consumption spending. Each has the potential to strengthen the denominator of the debt ratio over medium-term horizons through channels — productivity diffusion, labor-force quality, and research commercialization — that operate with longer lags than conventional fiscal policy but tend to be more durable when sustained across investment cycles. The scenario analysis in Section 2 provides a rough quantitative anchor for the fiscal arithmetic: each 50 basis point improvement in trend growth reduces the primary surplus required for debt stabilization by approximately 0.5 percentage points of GDP — roughly \$145 billion annually at current nominal GDP. Well-targeted productive-capacity investment that contributes even a fraction of that improvement over a decade would generate fiscal returns that offset a meaningful portion of its near-term cost, provided the composition

is weighted toward sectors with documented spillover characteristics — and, per Section 4.2, toward the domestically sited segments of those sectors specifically — rather than undifferentiated public spending.

The geopolitical-fragility discount developed in Section 4 carries a further implication for this composition choice. Domestic capital allocation toward advanced semiconductor fabrication addresses the revenue-capture gap identified in Section 4.2, but it does not, by itself, address the chokepoint exposure identified in Section 4.3, which is a function of where the rest of the world's fabrication capacity sits, not of how much capacity the United States adds to its own side of the ledger. Productive-capacity investment in this category is therefore necessarily paired with, rather than a substitute for, sustained diplomatic and security investment in the alliance architecture — with Taiwan, the Netherlands, Japan, and South Korea specifically — that keeps the rest of the matrix's indispensable capacity accessible. A reshoring strategy that treats the geopolitical dimension as exogenous to the investment decision will misprice the risk-adjusted return on that investment, in the same way that Table 8 shows the raw PI Index misprices it relative to the risk-adjusted ranking.

## 6.4 Debt Management Strategy

Maturity extension on outstanding debt can reduce rollover risk and smooth financing conditions independently of the primary balance. This lever is available to the Treasury as a policy tool and directly relevant to supervisory analysis: institutions holding short-duration Treasuries are less exposed to repricing risk than those holding long duration, per the duration mechanics quantified in Section 5.4 — a relevance that extends to the foreign central banks and commercial banks discussed in Section 3.1, whose own Treasury portfolio duration affects how much of the global collateral channel a given term-premium shock activates. A more actively managed maturity profile reduces the interest-rate shock transmission channel discussed in Section 5.1 and buys time for the structural adjustments in Sections 6.1 through 6.3 to take effect. It does not reduce the total stock of debt; it reduces the financial-stability risk of that stock under adverse rate scenarios, domestically and abroad.

The political economy of this mixed strategy deserves acknowledgment. Fiscal consolidation and productive investment can appear to pull in opposite directions over short budget horizons: consolidation requires reducing near-term spending, while productive investment requires sustaining or expanding certain categories of it. The tension is real but manageable if both tracks are designed with explicit attention to timing and composition. Consolidation concentrated in mandatory program reform and revenue adequacy preserves space for productive investment; productive investment targeted at sectors with measurable spillover characteristics minimizes the near-term fiscal cost of the growth strategy. Institutional continuity — specifically, the insulation of long-horizon research programs from annual appropriations volatility — is itself a form of policy design that improves the expected return on productive-capacity investment.

Table 14. Policy Channels, Expected Effects, and Supervisory Relevance

| Policy Channel                    | Expected Macro Effect                                                                                                                                        | Supervisory Relevance                                                                                                    |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Entitlement / health-care reform  | Slower mandatory spending growth; lower long-run primary deficit                                                                                             | Reduces sovereign-risk tail events; lowers long-run yield pressure, domestically and on foreign-held Treasury collateral |
| Broad revenue adjustment          | Higher receipts; improved primary balance                                                                                                                    | Supports debt stabilization and market confidence in fiscal trajectory, including foreign official demand                |
| Productivity-enhancing investment | Higher trend growth; improved r–g differential; broader tax base, weighted toward domestically sited, geopolitically resilient activity per Sections 4.2–4.3 | Eases medium-run pressure on bank portfolios; supports sovereign credit quality                                          |
| Debt maturity extension           | Lower rollover risk; smoother near-term financing profile                                                                                                    | Reduces duration-shock transmission quantified in Table 11, domestically and via the foreign holdings in Table 4         |
| STEM talent / immigration policy  | Stronger frontier labor supply; higher potential growth                                                                                                      | Supports productive-capacity conditions underlying long-run debt sustainability                                          |

| Policy Channel                                                | Expected Macro Effect                                                     | Supervisory Relevance                                                                                                           |
|---------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Alliance management (Taiwan, Netherlands, Japan, South Korea) | Sustains access to geopolitically fragile but indispensable supply chains | Determines whether Section 4's productive-indispensability mechanism is structurally reliable or merely conditionally available |

Source: Synthesis drawing on Bernanke (2011), CBO (2026), BIS (2025), IMF (2018), and Sections 3.1, 4, and 5.2 of this paper.

## 7. Limitations

No new structural model is estimated, and no specific productivity initiative is claimed to produce a guaranteed macro payoff. The  $r$ - $g$  differential is endogenous, as noted above, and the productive-capacity framework should be read as a structure for thinking about long-run fiscal resilience rather than a substitute for primary-balance arithmetic.

There is also a supervisory boundary to respect. The Federal Reserve does not set fiscal policy and should not be portrayed as responsible for closing the budget gap. Its role is to evaluate how fiscal developments affect financial institutions, markets, and the transmission of monetary policy. The analysis here stays within that mandate: it identifies fiscal conditions that raise supervisory risk, maps the channels through which that risk propagates domestically and across borders, and points to the policy complements — including productive investment and alliance management — that would improve those conditions. It does not recommend a specific fiscal package, and it does not take a position on the substance of any specific bilateral trade or export-control negotiation; it treats the existence and fragility of those negotiations as a supervisory input, not a policy recommendation.

The historical comparisons in Section 4 deserve explicit qualification. Britain's nineteenth-century debt reduction and the Dutch Republic's sovereign default in 1810 involved monetary structures, institutional arrangements, and geopolitical contexts materially different from those of a modern fiat reserve currency issuer. They are cited because they are consistent with the basic debt-arithmetic proposition that trend growth is a first-order determinant of debt trajectory; they are not presented as structurally parallel cases. Applying the specific debt ratios or timelines from those episodes to the United States would be inappropriate.

The counterarguments to the productive-capacity thesis also deserve acknowledgment. Gordon's work on the innovation slowdown presents serious empirical evidence that transformative productivity gains may be behind us and that current frontier technologies may be less economically powerful than those of earlier industrial eras (Gordon, 2016). Summers's secular stagnation thesis makes a related structural argument about persistent demand shortfalls and low equilibrium interest rates (Summers, 2014). If either analysis is substantially correct, the growth-driven component of debt stabilization is less available than the framework assumes, and the weight of adjustment falls more heavily on primary balances. That is a real limitation and an argument for urgency on both tracks rather than sequencing one behind the other.

A further limitation involves the endogeneity of productive investment timing. Sustained public investment in R&D, infrastructure, and education raises near-term spending, which adds to debt before the productivity payoff materializes. The long payoff horizon means that the fiscal case for productive investment is sensitive to discount rates: at sufficiently high discount rates, the net present value of growth returns may not justify the near-term addition to debt. The framework does not resolve this uncertainty; it argues that the expected return is positive and that the composition of investment matters — frontier research with measurable spillover characteristics carries higher expected fiscal returns than undifferentiated public spending, and, per Section 4.2, this composition effect is strongest for domestically sited activity specifically. That claim should be held with appropriate epistemic humility.

Blanchard's (2019) finding that  $r < g$  has historically been the norm in advanced economies provides a partial offset to the Gordon–Summers pessimism: if trend growth can be sustained, fiscal dynamics may be more benign than CBO's baseline interest assumptions imply. But that result strengthens rather than weakens the productive-capacity argument, since maintaining  $g$  above  $r$  requires exactly the structural conditions this paper describes.

The quantitative framework introduced in this revision sharpens several arguments but introduces its own estimation uncertainty, which should be stated with the same discipline applied above to the qualitative framework. The system-wide average securities duration used in Section 5.4's aggregate stress estimate (4.5 years) is an illustrative assumption pending FFIEC Call Report / FR Y-9C verification by asset-size cohort. The CDMO-biologics and defense-prime concentration scores in Table 7 are directional judgments based on general industry structure rather than calculated HHI values, pending BioPlan Associates / Citeline and SIPRI / Jane's primary-source data respectively. The hypothetical balance sheet in Table 11 is constructed for transmission mapping and does not represent any specific institution's filings.

The cross-border extensions added in this revision carry comparable caveats. The Geopolitical Fragility Discount in Section 4.1 is a structured analytical judgment about conflict probability and export-licensing stability, not a market-derived or formally estimated probability; reasonable analysts could score Taiwan Strait or Dutch export-licensing risk differently, and the discount factor of  $1 - \text{GFD}/20$  is a transparent but unvalidated functional form chosen for interpretability rather than estimated from data. The FBO cross-border transmission channel in Section 5.2 is supported by a well-documented historical precedent (the 2011 European episode) but the current aggregate scale of FBO exposure — last precisely quantified in this paper using a 2013 reserves figure — requires updating from current FFIEC 002 filings before being cited as a present-day exposure estimate. The global collateral and cross-currency basis channel in Section 3.1 is supported by an extensive academic and central-bank literature on the mechanism itself, but this paper does not separately estimate the basis-widening response to a U.S.-specific fiscal shock as distinct from the dollar-funding-stress episodes (2008, 2020) on which the cited literature is based; the directional logic is well established, the U.S.-fiscal-specific magnitude is not. None of these inputs should be cited as verified Federal Reserve or supervisory data without the underlying primary-source confirmation noted here; the value of the framework lies in the transmission logic it makes explicit, not in the precision of every illustrative input.

## 8. Conclusion

The United States is not facing an immediate debt crisis, but it is on a fiscal trajectory that compounds the financial-stability risks that supervisors monitor — and, this revision has argued, that other countries' financial systems monitor as well, whether or not their own supervisors frame it in those terms. CBO's baseline projects a decade of rising debt ratios and increasing net interest costs, driven by mandatory spending growth and a revenue structure that has not kept pace with federal obligations. Three structural headwinds — reserve currency costs, security expenditure asymmetries, and deindustrialization — compound that trajectory in ways that standard fiscal analysis does not always capture, and that are not fully addressable by conventional consolidation tools alone. This revision has shown that the deindustrialization channel can be expressed as a bounded tax-elasticity coefficient rather than only a directional claim, that the reserve-currency channel maps formally onto the financial-account side of the balance-of-payments identity and onward into a global collateral and cross-currency-basis mechanism, and that the productive-indispensability sectors most often invoked in this debate divide sharply along two axes that the raw engineering case for reshoring does not capture: whether the activity is domestically sited, and whether its indispensability is geopolitically conditional rather than structural.

The central supervisory lesson is that sovereign debt dynamics are not solely a matter for budget committees, and not solely a matter for U.S. supervisors. They shape Treasury-market functioning, term premia, bank valuation exposure, funding conditions, and the policy space available when the next downturn arrives — and, through the foreign banking organizations and global collateral channels developed in Sections 3.1 and 5.2, they shape the same variables for financial systems in Europe and Asia that depend on the dollar and on Treasury securities to fund themselves. The Federal Reserve's own Financial Stability Report has explicitly identified the domestic version of this risk channel; this paper's contribution is to show that the channel does not stop at the U.S. border. Section 5.4 has shown that the domestic channel can be calibrated quantitatively against the 2022–2023 regional banking episode, that its duration-driven and funding-driven components operate on different clocks, and that both have a cross-border mirror — in foreign central bank and commercial bank

Treasury holdings, and in the funding decisions of foreign banking organizations — that can transmit a domestic fiscal-stress event into an international one within the same episode rather than as a delayed second-order effect.

Durable debt stabilization therefore functions as more than a domestic fiscal exercise. It requires credible primary-balance adjustment, policies that sustain trend growth strong enough to keep  $r-g$  from widening further, and — as Section 4's risk-adjusted matrix makes explicit — sustained management of the alliance relationships on which the geopolitically concentrated portion of U.S. productive indispensability depends. The first reduces the numerator of the debt ratio; the second strengthens the denominator; the third determines whether the denominator's strength is durable or contingent on conditions outside U.S. control. Fiscal consolidation without productive capacity risks contracting the economy faster than the debt. Productive investment without fiscal discipline risks compounding near-term obligations while waiting for long-cycle returns. Productive investment that ignores geopolitical fragility risks mispricing its own expected return. And all three, conducted without attention to the global collateral and funding channels this revision has detailed, risk treating as a domestic balance-sheet problem what is, in fact, the foundational anchor of global dollar liquidity and, through it, of global financial stability. Managed as complements, and grounded in institutional continuity across political cycles, fiscal discipline, productive investment, and alliance management together represent the most defensible path to maintaining monetary and financial stability over the long run — for the United States, and for the financial systems that depend on it. None of the three tracks is optional; none is sufficient alone.

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